Curriculum Vitæ

Alexandre Bousse

1 Personal Details

First Name:	Alexandre
Family Name:	Bousse
Date of Birth:	13th of June, 1980
Place of Birth:	Rennes, France
Citizenship:	French
Position:	Full Researcher (*chargé de recherche*)
Laboratory:	*Laboratoire de Traitement de l'Information Médicale* (LaTIM) INSERM, UMR 1101, Brest, France
Email Address:	`alexandre.bousse@univ-brest.fr`
Homepage:	`https://abousse.github.io/`

2 Academic Career

2.1 Professional Experience

2021–present	**Full Researcher** (*chargé de recherche*), LaTIM, INSERM, UMR 1101, *Université de Bretagne Occidentale* (UBO), Brest, France
2018–2021	**Associate Professor**, LaTIM, INSERM, UMR 1101, UBO, Brest, France
2009–2018	**Post-doctorate**, Insitute of Nuclear Medicine (INM), University College London (UCL), London, UK
2005–2008	**PhD Candidate**, *Laboratoire du Traitement du Signal et de l'Image* (LTSI), INSERM, UMR 1099, *Université de Rennes 1*, Rennes, France

2.2 Education

2019	***Habilitation à diriger des recherches* (habilitation thesis)**, LaTIM, INSERM, UMR 1101, UBO, Brest, France Title: "*Contributions à la reconstruction tomographique compensée en mouvement*" Viva: 07/10/2019

Jury:
- Dimitris Visvikis, LaTIM, INSERM, UMR 1101, UBO, Brest, France
- Françoise Pène, UBO, Brest, France
- Claude Comtat, CEA, Orsay, France
- Andrew Reader, King's College London, London, UK
- Michel Defrise, *Université Libre de Bruxelles*, Brussels, Belgium

2005–2008 **PhD**, Signal Processing, LTSI, UMR 1099, *Université de Rennes 1*, Rennes, France, and Laboratory of Image Science and Technology (LIST), Southeast University (SEU), Nanjing, China
Title: "Inverse Problems and Application to Motion-Compensated Rotational X-ray Angiography"
Viva: 08/12/2008
Jury:
- Directors:
  - J.-L. Coatrieux, LTSI, INSERM, UMR 1099, *Université de Rennes 1*, Rennes, France
  - H. Shu, LIST, SEU, Nanjing, China
  - C. Toumoulin, LTSI, INSERM, UMR 1099, *Université de Rennes 1*, Rennes, France
- President: J. Demongeot, *Université Joseph Fourier*, Grenoble, France
- Referees:
  - J. Yang, Nanjing University of Science and Technology, Nanjing, China
  - C. Roux, *Télécom Bretagne*, Brest, France
- Reviewers:
  - L. Luo, LIST, SEU, Nanjing, China
  - D. Xia, Nanjing University of Science and Technology, Nanjing, China

2004–2005 **Master of Science (DEA), Statistics**, *Université de Rennes 1*, Rennes, France

2003–2004 **Master of Science (DESS), Statistics**, *Université de Rennes 1*, Rennes, France

1998–2003 **Bachelor of Science, Mathematics**, *Université de Rennes 1*, Rennes, France

3 Teaching Activities and Research Supervision

3.1 Teaching

2025–present **École d'Ingénieur Statistique Data Science et Big Data (ENSAI)** (last year), Bruz, France
Machine and deep learning
9-hour lecture per year.

2018–present **Institut Mines-Télécom (IMT) Atlantique** (2nd year), Brest, France
Advanced Medical Image Reconstruction
5-hour lecture per year & 6-hour lab per year
Mathematics and algorithms for image reconstruction
Machine and deep learning.

2018–2021 **M2 *Master Signaux Images en Biologie et Médecine* (SIBM), UBO**, Brest, France
Introduction to Image Reconstruction
6-hour lecture per year (ended in 2021)

	France: 12-hour lecture & 6-hour lab per year
2018–2022	**M2 *physique et instrumentation***, UBO, Brest Introduction to Image Reconstruction 12-hour lecture & 6-hour lab per year. (ended in 2022)
2018–2021	**Coordinator** of M2 SIBM at UBO
2018–2021	**M1 biologie-santé, UBO**, Brest, France Image Segmentation 6-hours lecture per year.
2018–2021	**Medical and Dental School**, UBO, Brest, France Pix 100 hours lab per year.
2004–2007	**ENSAI** (first year), Bruz, France Probability & Statistics (contractual teaching during PhD) Teaching topics: measure theory, random variables, parametric statistics, statistical hypothesis testing, linear regression 64-hour science lab in total.

3.2 PhD Supervision and Co-supervision

2024–present	**Clémentine Phung-Ngoc**—UBO **Title**: "Motion-corrected Low-dose PET/CT with Unsupervised Learning" **Director**: Alexandre Bousse (50%) **Co-director**: Olivier Saut (25%), Hong-Phuong Dang **Publications**: [J5], [C3] **Status**: ongoing
2024–present	**Antoine De Paepe**—UBO **Title**: "Motion-compensated CT Reconstruction with Unsupervised Learning" **Director**: Alexandre Bousse (100%) **Publications**: [J1], [C2] **Status**: ongoing
2023–present	**Thore Dassow**—UBO **Title**: "Scored-Based Diffusion Models for Material Decomposition in PCCT" **Director**: Alexandre Bousse (100%) **Publications**: [C1] (best Fully3D 2025 paper award) **Status**: ongoing
2022–present	**Corentin Vazia**—*Université Bretagne Sud* (UBS) **Title**: "Scored-Based Diffusion Models in Spectral CT" **Director**: Jacques Froment (50%) **Co-director**: Alexandre Bousse (50%) **Publications**: [J7], [C9], [C10] **Status**: ongoing
2021–2025	**Juan José Molina**—Pontificia Universidad Católica de Chile (PUC) **Title**: "Deep Learning Techniques for T_2^* Image Reconstruction" **Director**: Matias Courdurier (80%) **Co-supervisor**: Alexandre Bousse (20%) **Publications**: [J13] **Status**: completed

2021–2025	**Youness Mellak**—UBO **Title**: "Deep Learning Techniques for 3-gamma PET Imaging" **Director**: Dimitris Visvikis (25%) **Co-director**: Alexandre Bousse (75%) **Publications**: [J3], [J4], [C6]–[C8], [C16] **Status**: completed
2021–2025	**Valentin Gautier**—Institut National des Sciences Appliquées (INSA) Lyon **Title**: "*Reconstruction bimodale d'images TEP/IRM assisstée par intelligence artificielle*" **Director**: Bruno Sixou (50%) **Co-supervisor**: Alexandre Bousse (25%), Voichita Maxim (25%) **Publications**: [J12], [C5], [C12] **Status**: completed
2021–2024	**Noel Pinton**—UBO **Title**: "Synergistic PET/CT Reconstruction using Deep Learning" **Director**: Alexandre Bousse (100%) **Publications**: [J6], [C13], [C14] **Status**: completed
2021–2024	**Zhihan Wang**—UBO **Title**: "Spectral computed tomographic image reconstruction using deep learning" **Director**: Alexandre Bousse (75%) **Co-director**: Frank Vermet (25%) **Publications**: [C4], [J14], [C17] **Status**: completed
2017–2022	**Baptiste Laurent**—UBO **Title**: "*Estimation des diffusés en TEP par apprentissage profond*" **Director**: Nicolas Boussion **Co-director**: Alexandre Bousse (100%) **Publications**: [J2], [J15], [C23] **Status**: completed
2018–2022	**Sai Sundar Kandarpa**—UBO **Title**: "Tomographic Image Reconstruction with Direct Neural Network Approaches" **Director**: Alexandre Bousse (100%) **Publications**: [J10], [J11], [C15], [J16], [J21], [C20], [C29] **Status**: completed
2018–2022	**Suxer Alfonso Garcia**—UBO **Title**: "Multi-channel Computed Tomographic Image Reconstruction by Exploiting Structural Similarities" **Director**: Alexandre Bousse (100%) **Publications**: [J18], [C18], [C19], [C21] **Status**: completed
2017–2020	**Debora Giovagnoli**—IMT **Title**: "3-γ Image Reconstruction using LXe Compton Camera XEMIS2" **Director**: Dimitris Visvikis **Co-director**: Alexandre Bousse (50%) **Publications**: [J20] **Status**: completed

2016–2020	**Ludovica Brusaferri**—UCL **Title**: "Improving Quantification in non-TOF 3D PET/MR by Incorporating Photon Energy Information" **Director**: Kris Thielemans (50%) **Co-supervisor**: Alexnadre Bousse (50%) **Publications**: [J19], [J24], [C22], [C32], [C35] **Status**: completed
2016–2020	**Élise Émond**—UCL **Title**: "Improving Quantification in Lung PET/CT for the Evaluation of Disease Progression and Treatment Effectiveness" **Director**: Kris Thielemans (50%) **Co-supervisor**: Alexnadre Bousse (50%) **Publications**: [J25], [J26], [C25]–[C27] Status: completed
2014–2018	**Yu-Jung Tsai**—UCL **Title**: "Penalised Image Reconstruction Algorithms for Efficient and Consistent Quantification in Emission Tomography" **Director**: Kris Thielemans (50%) **Co-supervisor**: Alexnadre Bousse (50%) **Publications**: [J22], [J27], [C33], [J29], [C36], [C39], [C42] Status: completed
2010–2015	**Sarah Cade**—UCL **Title**: "Attenuation Correction of Myocardial Perfusion Scintigraphy Images without Transmission Scanning" **Director**: Brian F. Hutton (50%) **Co-supervisor**: Alexnadre Bousse (50%) **Publications**: [C58] **Status**: completed

3.3 Master Students Supervision

2024	Antoine De Paepe, Clementine Phung-Ngoc, Apolline Guerineau (ENSAI) "Compressed-sensing for CT reconstruction"
2023	Tiban Dorel (IMT Atlantique) "CT reconstruction with unrolling architectures"
2022	Mariana Yuli Sato do Nascimento (IMT Atlantique) "PET/CT denoising with conditional GANs"
2019	Celia Boutalbi (Université de Bordeaux) "Direct material decomposition in dual-energy CT"

4 Grants and External Funding

4.1 Funding at Current Position (since 01/09/2018)

2025–present	**Projets Intra- et Inter-Instituts Brestois (IB) de l'UBO** (France) Amount awarded: €6,000 **Role: PI**

	Project Title: "Diffusion Spectrale"
2024–present	**Inserm-Inria PhD grant** (France) Amount awarded: €130,000 **Role: PI** Project Title: "Motion-corrected Low-dose PET/CT with Unsupervised Learning"
2024–present	**Région Bretagne / UBO PhD grant** (France) Amount awarded: €130,000 **Role: PI** Project Title: "Motion-compensated CT Reconstruction with Unsupervised Learning"
2023–present	**France Life Imaging WP4** (France) Amount awarded: €20,000 **Role: PI** Project Title: "Deepmulti: Deep Learning Techniques for Multimodal Image Reconstruction"
2020-2024	**ANR – ANR-20-CE45-0020** (France) Amount awarded: €496,800 **Role: PI** Project Title: "MultiRecon: Machine-Learning for Multimodal Medical Image Reconstruction"
2019–2022	**France Life Imaging WP4** (France) Amount awarded: €24,000 **Role: PI** Project Title: "Dual-Tracer in Dynamic PET" (complement funding to the *Émergence* project)
2019–2022	**AO *Émergence Cancéropôle Grand Ouest*** (France) Amount awarded: €15,000 **Role: PI** Project Title: "Dual-Tracer in Dynamic PET"

4.2 Funding during Postdoc at UCL (2009–2018)

2016–2018	**GE Healthcare** (USA) Amount awarded: $150,000 **Role: co-PI** PI: Kris Thielemans Project Title: "Motion-Compensated PET/CT"
2013–2015	**FP7 – HEALTH Program 305311** (EU) Amount awarded: €575,622 (€5,981,463 in total for all the EU partners) **Role: Research Fellow** PI: Brian F. Hutton Project Title: "Development of an integrated SPECT/MRI system"
2013–2016	**EPSRC – EP/K005278/1** (UK) Amount awarded: £1,274,298 **Role: WP leader** PI: Brian F. Hutton Project Title: "Exploiting the Unique Quantitative Capabilities Offered by Simultaneous PET/MRI"
2009–2012	**EPSRC – EP/G026483/1** (UK)

Amount awarded: £767,088
Role: Research Fellow
PI: Brian F. Hutton
Project Title: "Optimising Reconstruction to Accommodate Complex System Models for Emission Tomography"

5 Academic Service and Scientific Diffusion

5.1 Meetings Organisation

5.1.1 Conferences

2022–present	Member of IEEE Nuclear and Medical Imaging Sciences Council
2023	**Session Chairman**, IEEE Medical Imaging Conference, Vancouver, Canada
2022	**Session Chairman**, IEEE Medical Imaging Conference, Milan, Italy
2021	**Student Paper Awards Committee**, International Conference on Fully Three-Dimensional Image Reconstruction in Radiology and Nuclear Medicine
2018	**Session Chairman**, IEEE Medical Imaging Conference, Sydney, Australia

5.1.2 Workshops

2022	AI Wild West (organiser) https://wildwestworkshop.github.io/
2022	Rencontre Lyonnaise en Imagerie d'Émission (organiser) https://emilyworkshop.github.io/

5.2 Scientific Evaluation

5.2.1 Peer Reviewing

Journal Associate Editor

- IEEE TRPMS
- Frontiers in Medicine

Journal Peer Review

- IEEE Transactions on Medical Imaging
- IEEE Transactions on Biomedical Engineering
- IEEE Transactions on Radiation and Plasma Medical Sciences
- IEEE Transactions on Computational Imaging
- Physics in Medicine and Biology
- Neuroimage
- PLOS one
- Philosophical Transactions A
- Frontiers in Nuclear Medicine
- European Journal of Nuclear Medicine and Molecular Imaging

Grant Peer Review

- Nantes Excellence Trajectory (NEXT)
- *Wetenschappelijk Fonds Willy Gepts* (WFWG)
- Netherlands Organisation for Scientific Research (NWO)
- Expertise Scientifique Cifre

Conference Committee

- MICCAI 2020, 2021, 2022, 2023, 2024, 2025
- Fully 3D 2021, 2023, 2025
- IEEE Nuclear Science Symposium and Medical Imaging Conference 2014, 2015, 2016, 2017, 2018, 2019, 2021, 2022, 2023, 2024

5.2.2 PhD Examination

2023	**Kannara Mom**, "Deep learning-based phase retrieval for X-ray phase contrast imaging", INSA Lyon, Lyon, France.
2023	**Guillaume Corda**, "Joint PET-MR deep learned reconstruction", King's College London, London, UK.
2022	**Louise Friot**, "*Méthodes de reconstruction avancées en tomographie dentaire par faisceau conique*", INSA Lyon, Lyon, France.
2021	**Zacharias Chalampalakis**, "Modelling and Reconstruction of Whole-Body Parametric. Maps in PET-MRI Pharmacological Imaging", Biomaps, Orsay, France.

5.2.3 PhD Jury

2024	**Noel Pinton**, "Synergistic PET/CT Reconstruction using Deep Learning", UBO, Brest, France.
2024	**Zhihan Wang**, "Spectral computed tomographic image reconstruction using deep learning", UBO, Brest, France.
2022	**Suxer Alfonso Garcia**, "Multi-channel computed tomographic image reconstruction by exploiting structural similarities", UBO, Brest, France.
2022	**Sai Sundar Kandarpa**, "Tomographic image reconstruction with direct neural network approaches", UBO, Brest, France.
2020	**Debora Giovagnoli**, "3-γ Image Reconstruction using LXe Compton Camera XEMIS2", UBO, Brest, France.

5.2.4 Other Jurys

2022–present	Jury contrat doctoral établissement (CDE), école doctorale biologie santé, UBO.

5.3 Recent Invited Talks

Jul. 2024	"*Tomodensitométrie à comptage photonique : projets du LaTIM en reconstruction d'image*", LaTIM Day , LaTIM, Brest, France.

Nov. 2023	“*Modèles génératifs pour reconstruction multimodale*”, GDR ISIS, INSA, Lyon, France.
Mar. 2022	“Generative Models for Multichannel Image Reconstruction”, *Pontificia Universidad Católica de Chile*, Santiago, Chile.
Oct. 2021	“Multirecon: Machine Learning for Multimodal Medical Image Reconstruction”, INSA, Lyon, France.
Jul. 2019	“Innovations in Image Reconstruction”, LaTIM, Brest, France.
Feb. 2019	“*Reconstruction d’image en tomographie à émission de positons par maximum de vraisemblance avec compensation du mouvement respiratoire* ”, *Laboratoire de Mathématiques de Bretagne Atlantique*, Brest, France.
Jun. 2018	“Respiratory Motion Correction in PET/CT and PET/MR”, Mathematical Methods for Spatiotemporal Imaging, SIAM Conference on Imaging Science 2018, Bologna, Italy.
Mar. 2017	“Maximum-Likelihood PET Reconstruction and Motion Estimation”, *Pontificia Universidad Católica de Chile*, Santiago, Chile.
Sept. 2016	“Direct Motion Compensation in Attenuation-Corrected PET/CT and PET/MR Reconstruction”, UCL PET/MR Methods Symposium, London, UK.
May 2016	“*Reconstruction en PET-CT avec compensation du mouvement par techniques de maximum de vraisemblance*”, CEA, Orsay, France.
Mar. 2016	“Motion-Compensated PET Image Reconstruction by Maximum-Likelihood”, Newton Project Workshop on Brazil/UK Collaboration: the Future of Molecular Imaging, Recife, Brazil.
Nov. 2015	“Gated PET Reconstruction with Motion Compensation and Attenuation Correction using non-Gated CT”, Brain Institute, *Hospital Israelita Albert Einstein*, São Paulo, Brazil.

5.4 Software Development: JRM

Name	Joint Reconstruction and Motion estimation (JRM)
Language	Matlab/C++
Description	Joint Reconstruction and Motion estimation (JRM) is a toolbox for motion-compensated attenuation-corrected PET reconstruction that I developed for UCL and GE Healthcare. While the full version cannot be distributed, a “light” version is available at the address below.
Source code	https://gitlab.com/abousse/jrm_lite

6 Publications

Journal articles & preprints

[J1] A. De Paepe, **A. Bousse**, C. Phung-Ngoc, Y. Mellak, and D. Visvikis, “Adaptive diffusion models for motion-corrected cone-beam head CT,” *arXiv preprint arXiv:2504.14033*, 2025. [Online]. Available: https://arxiv.org/abs/2504.14033.

[J2] B. Laurent, **A. Bousse**, T. Merlin, A. Rominger, K. Shi, and D. Visvikis, “Evaluation of deep learning-based scatter correction on a long-axial field-of-view PET scanner,” *European journal of nuclear medicine and molecular imaging*, 2025. DOI: 10.1007/s00259-025-07120-6. [Online]. Available: https://arxiv.org/pdf/2501.01341.

[J3] Y. Mellak, **A. Bousse**, T. Merlin, É. Émond, and D. Visvikis, “Dual-input dynamic convolution for positron range correction in PET image reconstruction,” *arXiv preprint arXiv:2503.00587*, 2025. [Online]. Available: https://arxiv.org/abs/2503.00587.

[J4] Y. Mellak, **A. Bousse**, T. Merlin, D. Giovagnoli, and D. Visvikis, “Direct3γpet: A pipeline for direct three-gamma PET image reconstruction,” *IEEE Transactions on Radiation and Plasma Medical Sciences*, 2025. DOI: 10.1109/TRPMS.2025.3577810. [Online]. Available: https://arxiv.org/pdf/2407.18337.

[J5] C. Phung-Ngoc, **A. Bousse**, A. De Paepe, H.-P. Dang, O. Saut, and D. Visvikis, “Joint reconstruction of activity and attenuation in PET by diffusion posterior sampling in wavelet coefficient space,” *arXiv preprint arXiv:2505.18782*, 2025. [Online]. Available: https://arxiv.org/abs/2505.18782.

[J6] N. J. Pinton, **A. Bousse**, C. Cheze-Le Rest, and D. Visvikisc, “Multi-branch generative models for multichannel imaging with an application to PET/CT synergistic reconstruction,” *IEEE Transactions on Radiation and Plasma Medical Sciences*, vol. 9, no. 5, pp. 654–666, 2025. DOI: 10.1109/TRPMS.2025.3532176. [Online]. Available: https://arxiv.org/abs/2404.08748.

[J7] C. Vazia, T. Dassow, **A. Bousse**, J. Froment, B. Vedel, F. Vermet, A. Perelli, J.-P. Tasu, and D. Visvikis, “Material decomposition in photon-counting computed tomography with diffusion models: Comparative study and hybridization with variational regularizers,” *arXiv preprint arXiv:2503.15383*, 2025. [Online]. Available: https://arxiv.org/pdf/2503.15383.

[J8] H. Xu, **A. Bousse**, and A. Perelli, “Direct dual-energy CT material decomposition using model-based denoising diffusion model,” *arXiv preprint arXiv:2507.18012*, 2025. [Online]. Available: https://arxiv.org/pdf/2507.18012.

[J9] J. Zhang, **A. Bousse**, L. Imbert, S. Xue, K. Shi, and J. Bert, “Semi-supervised learning for dose prediction in targeted radionuclide: A synthetic data study,” *arXiv preprint arXiv:2503.05367*, 2025. [Online]. Available: https://arxiv.org/pdf/2503.05367.

[J10] **A. Bousse**, V. S. S. Kandarpa, S. Rit, A. Perelli, M. Li, G. Wang, J. Zhou, and G. Wang, “Systematic review on learning-based spectral CT,” *IEEE Transactions on Radiation and Plasma Medical Sciences*, vol. 8, no. 2, pp. 113–137, 2024. DOI: 10.1109/TRPMS.2023.3314131. [Online]. Available: https://arxiv.org/abs/2304.07588.

[J11] **A. Bousse**, V. S. S. Kandarpa, K. Shi, K. Gong, J. S. Lee, C. Liu, and D. Visvikis, “A review on low-dose emission tomography post-reconstruction denoising with neural network approaches,” *IEEE Transactions on Radiation and Plasma Medical Sciences*, vol. 8, no. 4, pp. 333–347, 2024. DOI: 10.1109/TRPMS.2023.3349194. [Online]. Available: https://arxiv.org/abs/2401.00232.

[J12] V. Gautier, **A. Bousse**, F. Sureau, C. Comtat, V. Maxim, and B. Sixou, “Bimodal PET/MRI generative reconstruction based on VAE architectures,” *Physics in Medicine & Biology*, vol. 69, 2024. DOI: 10.1088/1361-6560/ad9133. [Online]. Available: https://iopscience.iop.org/article/10.1088/1361-6560/ad9133.

[J13] J. Molina, **A. Bousse**, T. Catalán, Z. Wang, M. Petrache, F. Sahli, C. Prieto, and M. Courdurier, “CConnect: Synergistic convolutional regularization for cartesian T2* mapping,” *arXiv preprint arXiv:2404.18182*, 2024. [Online]. Available: https://arxiv.org/abs/2404.18182.

[J14] Z. Wang, **A. Bousse**, F. Vermet, J. Froment, B. Vedel, A. Perelli, J.-P. Tasu, and D. Visvikis, “Uconnect: Synergistic spectral CT reconstruction with U-Nets connecting the energy bins,” *IEEE Transactions on Radiation and Plasma Medical Sciences*, vol. 8, no. 2, pp. 222–233, 2024. DOI: 10.1109/TRPMS.2023.3330045. [Online]. Available: https://arxiv.org/abs/2311.00666.

[J15] B. Laurent, **A. Bousse**, T. Merlin, S. Nekolla, and D. Visvikis, "PET scatter estimation using deep learning U-net architecture," *Physics in Medicine & Biology*, vol. 68, no. 6, p. 065 004, 2023. DOI: 10.1088/1361-6560/ac9a97.

[J16] V. S. S. Kandarpa, A. Perelli, **A. Bousse**, and D. Visvikis, "LRR-CED: Low-resolution reconstruction-aware convolutional encoder–decoder network for direct sparse-view CT image reconstruction," *Physics in Medicine & Biology*, vol. 67, no. 15, p. 155 007, 2022. DOI: 10.1088/1361-6560/ac7bce.

[J17] F. Lamare, **A. Bousse**, K. Thielemans, C. Liu, T. Merlin, H. Fayad, and D. Visvikis, "PET respiratory motion correction: Quo vadis?" *Physics in Medicine & Biology*, vol. 67, no. 3, 03TR02, 2022. DOI: 10.1088/1361-6560/ac43fc. [Online]. Available: https://hal.science/hal-04090024.

[J18] A. Perelli, S. L. Alfonso Garcia, **A. Bousse**, J.-P. Tasu, N. Efthimiadis, and D. Visvikis, "Multi-channel convolutional analysis operator learning for dual-energy CT reconstruction," *Physics in Medicine & Biology*, vol. 67, no. 6, p. 065 001, 2022. DOI: 10.1088/1361-6560/ac4c32. [Online]. Available: https://arxiv.org/abs/2203.05968v1.

[J19] L. Brusaferri, É. C. Émond, **A. Bousse**, R. Twyman, A. C. Whitehead, D. Atkinson, S. Ourselin, B. F. Hutton, S. Arridge, and K. Thielemans, "Detection efficiency modelling and joint activity and attenuation reconstruction in non-TOF 3D PET from multiple-energy window data," *IEEE Transactions on Radiation and Plasma Medical Sciences*, vol. 6, no. 1, pp. 87–97, 2021. DOI: 10.1109/TRPMS.2021.3064239.

[J20] D. Giovagnoli, **A. Bousse**, N. Beaupere, C. Canot, J.-P. Cussonneau, S. Diglio, A. Iborra Carreres, J. Masbou, T. Merlin, E. Morteau, Y. Xing, Y. Zhu, D. Thers, and D. Visvikis, "A pseudo-TOF image reconstruction approach for three-gamma small animal imaging," *IEEE Transactions on Radiation and Plasma Medical Sciences*, vol. 5, no. 6, pp. 826–834, 2021. DOI: 10.1109/TRPMS.2020.3046409.

[J21] V. S. S. Kandarpa, **A. Bousse**, D. Benoit, and D. Visvikis, "DUG-RECON: A framework for direct image reconstruction using convolutional generative networks," *IEEE Transactions on Radiation and Plasma Medical Sciences*, vol. 5, no. 1, pp. 44–53, 2021. DOI: 10.1109/TRPMS.2020.3033172. [Online]. Available: https://arxiv.org/abs/2012.02000.

[J22] Y.-J. Tsai, **A. Bousse**, S. Arridge, C. W. Stearns, B. F. Hutton, and K. Thielemans, "Penalized PET/CT reconstruction algorithms with automatic realignment for anatomical priors," *IEEE Transactions on Radiation and Plasma Medical Sciences*, vol. 5, no. 3, pp. 362–372, 2021. DOI: 10.1109/TRPMS.2020.3025540. [Online]. Available: https://arxiv.org/abs/1911.08012.

[J23] **A. Bousse**, M. Courdurier, É. C. Émond, K. Thielemans, B. F. Hutton, P. Irarrazaval, and D. Visvikis, "PET reconstruction with non-negativity constraints in projection space: Optimization through hypo-convergence," *IEEE Transactions on Medical Imaging*, vol. 39, no. 1, pp. 75–86, 2020. DOI: 10.1109/TMI.2019.2920109. [Online]. Available: https://hal.archives-ouvertes.fr/hal-02144923.

[J24] L. Brusaferri, **A. Bousse**, É. C. Émond, R. Brown, Y.-J. Tsai, D. Atkinson, S. Ourselin, C. C. Watson, B. F. Hutton, S. Arridge, *et al.*, "Joint activity, attenuation and scatter estimation from multiple energy window data in non-TOF 3D PET: A preliminary study," *IEEE Transactions on Radiation and Plasma Medical Sciences*, vol. 4, no. 4, pp. 410–421, 2020. DOI: 10.1109/TRPMS.2020.2978449. [Online]. Available: https://ieeexplore.ieee.org/document/9024002.

[J25] É. C. Émond, **A. Bousse**, L. Brusaferri, B. F. Hutton, and K. Thielemans, "Improved PET/CT respiratory motion compensation by incorporating changes in lung density," *IEEE Transactions on Radiation and Plasma Medical Sciences*, vol. 4, no. 5, pp. 594–602, 2020. DOI: 10.1109/TRPMS.2020.3001094. [Online]. Available: https://ieeexplore.ieee.org/document/9112356.

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